

Week-1

1.

Machine Learning :— ML is the study of computer algorithms that improve automatically through experience and the use of the data.

→ Applications of ML:-

→ Predicting whether an email is spam or not.

→ Forecasting company's revenue for the upcoming year.

→ Recommending a product that a client may be interested in, based on past experience.

→ ML Model ↪

2. A machine learning model is a program that can find patterns or make decisions from a previously unseen dataset.

Types of Models:

↪ ① Predictive Models

↪ ② Probabilistic models.

Predictive Models

↪ Regression Model.

✓ Predicting the price of land, based on its area and distance from the market

→ Classification Models

↪ Predicting whether an email is spam or not.

↪ Predicting whether the patient is having cancer or not.

Probabilistic Models

→ 3. Probabilistic models are used for evaluating how likely an event or configuration is.

Learning Algorithm:-

Data $\rightarrow$ Models

eg: Price = $a \times (\text{area}) + b \times (\text{rooms})$

+ $c \times (\text{distance to metro})$

Parameters:- a, b, c

→ Algorithm use data to get the "best" parameters

Supervised ML: Regression

Given: $\{(x^1, y^1), (x^2, y^2), \dots, (x^n, y^n)\}$

Goal: find a model f such that $\underbrace{f(x^i) \approx y^i}$

Algorithm outputs a model $f: \mathbb{R}^d \rightarrow \mathbb{R}$

How to evaluate this model?

$$\text{Loss} = \frac{1}{n} \sum_{i=1}^n (f(x^i) - y^i)^2$$

$$f(x) = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b$$

$$= w^T x + b$$

$$= \sum_{j=1}^n w_j x_j + b$$

Example:-

5.

(3.4) ✓✓✓

x	y	$f(x) = (x^2 - 2)/5$
1	(2)	(-0.2)
3	4	1.4
4	3	2.8
5	6	4.6

$$f(x) = \frac{x^2 - 2}{5}$$

$$\text{Loss} = \frac{1}{n} \sum_{i=1}^n (f(x_i) - y_i)^2$$

$$= \frac{1}{4} [(-0.2 - 2)^2 + (1.4 - 4)^2 + (2.8 - 3)^2 + (4.6 - 6)^2]$$

$$= 3.4.$$

Supervised learning:- Classification

Given: $\{(\underline{x}^1, \underline{y}^1), (\underline{x}^2, \underline{y}^2), \dots, (\underline{x}^n, \underline{y}^n)\}$

here, $\underline{x}^i \in \mathbb{R}^d$ & $\underline{y}^i \in \{+1, -1\}$

Algorithm outputs a model $f: \mathbb{R}^d \rightarrow \{+1, -1\}$

↓
spam not
 spam

How to evaluate this model?

6.

$$\text{loss} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(f(x^i) \neq y^i)$$

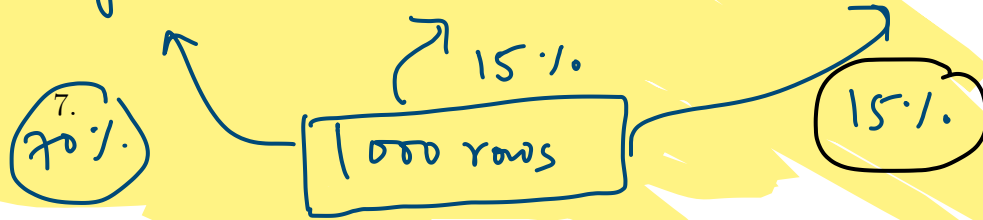
$$f(x) = \text{sign}(w^T x + b)$$

Eg:-

x	y	$f(x) = \text{sign}(x_1 - 3x_2)$
$[1, 0]$	$+1$	$+1$ ✓
$[2, 2]$	$+1$	-1
$[3, 1]$	$+1$	$+1$ ✓
$[4, 5]$	-1	-1 ✓
$[5, 4]$	-1	-1 ✓
$[1, 5]$	-1	-1 ✓

$$\text{loss} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(f(x^i) \neq y^i) = \frac{1}{6} = 0.166.$$

Training data, Validation data, Test data



Studying for an Exam

Stage

Training data: → you read textbooks & solve practice questions.
(what we learn from)

↓
learn the concepts.
(to understand the pattern)

Validation data → you take a mock test to see how well you're prepared.
(what we practice on)

to improve the model

Test data → you appear for the final exam.

to measure real performance.

(what we finally test on)

Unsupervised learning

8.

30 features

4 features

without losing
much information.

Understanding data

Build models that compress, explain & group data.

Eg:- 12 subjects.

$$s_1 \quad s_2 \quad s_3 \quad \dots \quad s_{12}$$

1

2

3

1

•

.

•

20,000

→ Too many dimensions

12 features

2 features $\rightarrow$ DS

→ frog

8

Page

bs

Dimensionality Reduction: -

9.

↳ Compression and Simplification.

Data: $\{x^1, x^2, \dots, x^n\}$

Encoder: $f: \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$; $d' < d$

Decoder: $g: \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$; $d' < d$

Goal: $g(f(x^i)) \approx x^i$

$$\text{Loss} = \frac{1}{n} \sum_{i=1}^n \|g(f(x^i)) - x^i\|^2$$

5-148

Eg:- encoder: $f(x) = x_1 - x_2 + 0.5x_3$, $g(u) = [0.3u, -4u, u]$

x	$f(x)$	$g(f(x))$	$\ g(f(x)) - x\ ^2$
$[-1, 1, -1]$	-2.5	$[-0.75, 2.5, -2.5]$	$(-0.75+1)^2 + (2.5-1)^2 + (-2.5+1)^2$ $= 4.56$
$[1, 2, 1]$	-0.5	$[-0.15, 0.5, -0.5]$	5.82
$[1, 1, -1]$	-0.5	$[-0.15, 0.5, -0.5]$	1.82
$[1, -1, 2]$	3	$[0.9, -3, 3]$	5.01
$[0, 1, 3]$	0.5	$[0.15, -0.5, 0.5]$	8.52

$$\text{loss} = \frac{1}{5} [4.56 + 5.82 + 1.82 + 5.01 + 8.52]$$

10.

$$= \underline{\underline{5.148}} \quad \checkmark$$

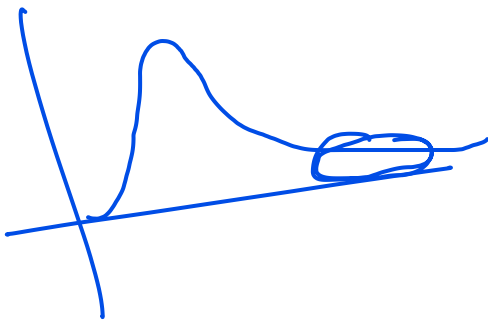
Unsupervised learning : Density Estimation.

↓
process of estimating the
prob. distn of a random variable
based on observed data.

Eg:- Fraud detecting in banking

↳ estimate the "normal" spending &
behaviour of a customer.

→ Transactions with low estimated density
(rare events) can be flagged as
suspicious.



Data: $\{x^1, x^2, \dots, x^n\}$

11. $x^i \in \mathbb{R}^d$

• probability mapping $p: \mathbb{R}^d \rightarrow \mathbb{R}_+$ that 'sums' to one.

Goal:- $p(x)$ is large if $x \in \text{Data}$, and low otherwise.

$$\underline{\text{loss}} := \frac{1}{n} \sum_{i=1}^n -\log(p(x^i))$$

eg:-

x	$p(x)$	$-\log(p(x))$
2.5	$1/5$	
1	$1/5$	
3	$1/5$	
4.5	$1/5$	
4.95	$1/5$	

$$\text{loss} = \frac{1}{5} \sum_{i=1}^5 -\log(p(x^i))$$

$$= 0.69.$$

Week-2

1. Continuity

A function $f(x)$ is continuous at $x=a$ if:

$$f(a) = \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

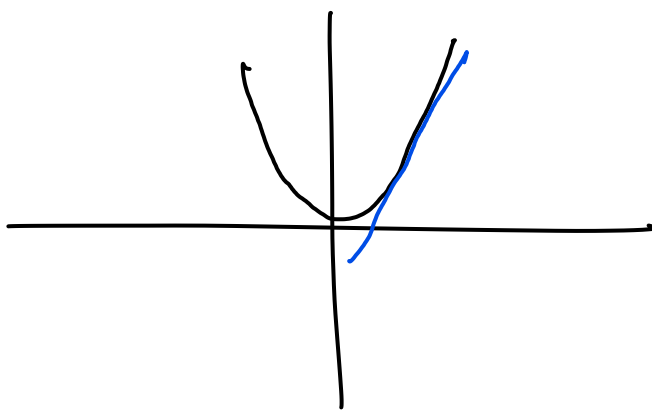
Differentiability! -

A function $f(x)$ is differentiable at $x=a$ if

$$f'(a) = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$$

↓

is the slope of the tangent to the curve $(x, f(x))$ at $x=a$.



$$f'(a) \approx \frac{f(x) - f(a)}{x - a}$$

12

$$\Rightarrow f'(a)(x-a) \approx f(x) - f(a) \Rightarrow$$

Linear Approximation :

13.

$$f(x) \approx f(a) + f'(a) \cdot (x-a)$$

$$L_a[f(x)] = f(a) + f'(a) \cdot (x-a).$$

eg:- $f(x) = \underline{3x+8}$ ✓ linear form

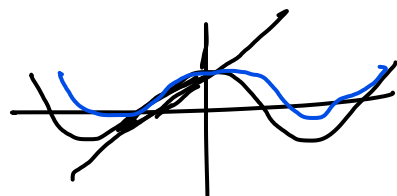
$$L_0[f(x)] = f(0) + f'(0) \cdot (x-0)$$

$$= 8 + 3x = \underline{3x+8} \text{ linear form.}$$

eqⁿ of a tangent line:

$$y - y_1 = m(x - x_1)$$

$$y = y_1 + m(x - x_1) \quad \checkmark$$



Taylor series :

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

Higher order approx

14. Linear approximation:-

$$L(x) = f(a) + f'(a) \cdot (x-a) \quad \checkmark \checkmark$$

Quadratic approximation:-

$$L(x) = f(a) + f'(a) \cdot (x-a) + \frac{f''(a)}{2!} (x-a)^2$$

Higher-order approximation:-

$$L(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \frac{f'''(a)}{3!} (x-a)^3 + \dots$$

Multivariate linear approximation:-

$$L(x) = \underbrace{f(a)} + \underbrace{f'(a)} (x-a) \rightarrow \text{single}$$

The linear approx. of a function f of two variables x and y in the neighborhood of (a, b) is:

$$L(x, y) = f(a, b) + \frac{\partial f}{\partial x}(a, b) \cdot (x - a)$$

15.

$$+ \frac{\partial f}{\partial y}(a, b) \cdot (y - b)$$

Directional Derivatives:-

→ $\frac{\partial f}{\partial x}(x, y)$ = Rate of change of f as we vary x
(keeping y fixed).

→ $\frac{\partial f}{\partial y}(x, y)$ = Rate of change of f as we vary y
(keeping x fixed)

• Directional derivative of $f(x, y)$ = Rate of change of f if we allow both x and y to change simultaneously (in some direction (u)).

$$\begin{aligned} D_{\vec{u}} f(x, y) &= \nabla f \cdot (\vec{u}) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] \cdot [u_1, u_2] \end{aligned}$$

15

$$= u_1 \cdot \frac{\partial f}{\partial x} + u_2 \cdot \frac{\partial f}{\partial y}$$

Contours (level curves) are obtained by slicing
the _{16.} surface with planes parallel to xy plane.

$$f(x, y) = x^2 + y^2$$
$$(x, y, f(x, y))$$

$$f(x, y) = 1$$

$$f(x, y) = 2$$

Week-③

- ^{17.} Four fundamental subspaces.
 - Orthogonal Vectors & Subspaces.
 - Projections
 - Least squares & Projection onto a Subspace
-

→ Four Fundamental Subspaces.

↳ Suppose A is a $m \times n$ matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m \times n}$$

- ① The colⁿ space is $C(A)$, a subspace of $\mathbb{R}^n$
- ② The Row space is $C(A^T) = R(A)$, a subspace of $\mathbb{R}^m$.
- ③ The null space is $N(A)$, a subspace of $\mathbb{R}^n$.
- ④ The left nullspace is $N(A^T)$, a subspace of $\mathbb{R}^m$.

$$\underline{Ax = b}.$$

18.

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}_{m \times n} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}_{n \times 1} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}_{m \times 1}$$

$$\Rightarrow \begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ \vdots & \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m. \end{aligned}$$

$$\Rightarrow \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} x_1 + \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} x_2 + \dots + \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} x_n = b.$$

→ linear combⁿ of columns of $A = C(A)$

19.

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix}$$

Linear combⁿ of rows of A.

$$A^T = \begin{bmatrix} 1 & 2 & -1 \\ 4 & 8 & -4 \\ 2 & 4 & -2 \end{bmatrix} \downarrow$$

$$= \underline{\underline{C(A^T)}}.$$

Nullspace of A:

$$Ax = 0.$$

$$N(A) = \{ x \in \mathbb{R}^n \mid Ax = 0 \}$$

Left Null space : $N(A^T)$

Set of all vectors $y \in \mathbb{R}^m$ such that

$$A^T y = 0$$

$$\Rightarrow \overbrace{(y^T A)} = 0^T$$

↑
left

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix}$$

20.

$$\text{rank}(A) = 1$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 + R_1$$

$$\begin{bmatrix} \boxed{1} & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{pivot}$$

$$C(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \right\}$$

$R(A)$ is same as $C(A^T)$

$$A^T = \begin{bmatrix} 1 & 2 & -1 \\ 4 & 8 & -4 \\ 2 & 4 & -2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 4R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} \boxed{1} & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{pivot}$$

20

$$\underline{\underline{R(A)}} = C(A^T) = \text{span} \left\{ \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \right\}.$$

21.

N(A): $Ax = 0$

$$\begin{bmatrix} 1 & 4 & 2 \\ 2 & 0 & 4 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\mathbb{R}$ $\begin{bmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\rightarrow x_1 + 4x_2 + 2x_3 = 0$$

x_2 & $x_3 \rightarrow$ free variables

if $x_2 = 1, x_3 = 0$

then $x_1 = -4$

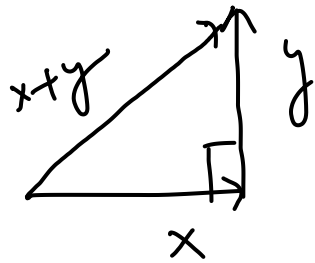
if $x_2 = 0, x_3 = 1$

then $x_1 = -2$

$$N(A) = \text{span} \left\{ \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Orthogonal Vectors :-

22.



→ Right-angled triangle

$$\underbrace{\|x\|^2}_{x^T x} + \underbrace{\|y\|^2}_{y^T y} = \|x+y\|^2$$
$$x^T x + y^T y = (x+y)^T (x+y)$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$
$$x^T x$$
$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + x_2^2$$

$$\Rightarrow x^T(x+y) = x^T x + x^T y + y^T x + y^T y$$

$$\Rightarrow 0 = 2x^T y$$

$$\Rightarrow \boxed{x^T y = 0}$$

eg:- $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, y = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, x+y = \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}$

$$\|x\|^2 = 14, \quad \|y\|^2 = 5, \quad \|x+y\|^2 = 19$$

22

$$\boxed{x^T y = 2 - 2 + 0 = 0.}$$

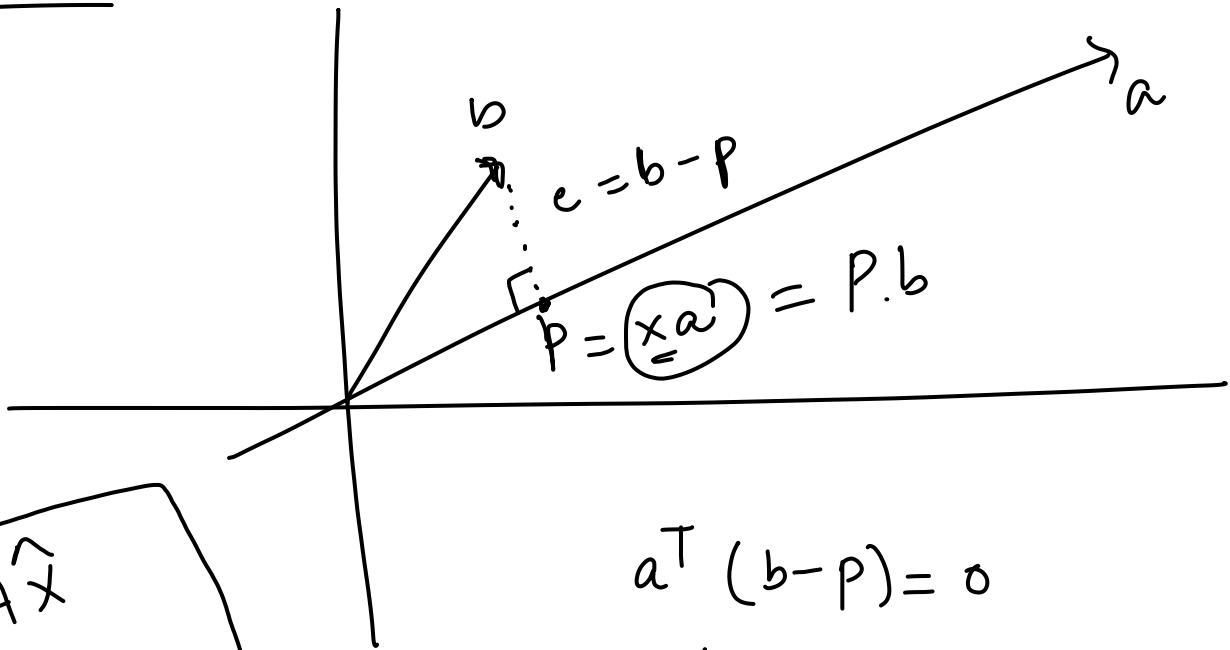
* Subspace S is orthogonal to subspace T means:
 $\rightarrow$ every vector in S is orthogonal to every vector in T .

23.

(*) Row space is orthogonal to Null space.

* Col space is _____ left Null space.

Projections:



$$\begin{aligned} p &= A \hat{x} \\ A^T(b - A \hat{x}) &= 0 \\ \downarrow \\ A^T b &= A^T A \hat{x} \end{aligned}$$

$$\begin{aligned} a^T(b - p) &= 0 \\ \Rightarrow a^T(b - x a) &= 0 \\ \Rightarrow a^T b - x a^T a &= 0 \end{aligned}$$

23

$$\Rightarrow x = \frac{a^T b}{a^T a}$$

$$p = a^T \rightarrow \frac{a a^T}{a^T a} \cdot b$$

24. $p =$

$$\text{proj}, b = P \cdot b$$

$$\Rightarrow P = \frac{a a^T}{a^T a}$$

Projection Matrix

$$\text{Ex } P^T = P \text{ and } P^2 = P$$

$AX = b$ $\rightarrow$ may have no solution.

$$A^T x = p$$

least squares Approximation:

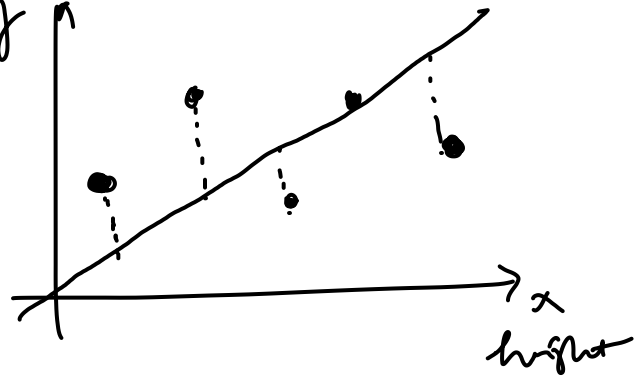
25.

→

$Ax=b$ has no solution.
too many equations
 A has more rows than
cols.

more equations than unknown.

weight



$$\left. \begin{array}{l} x+y=2 \\ x-y=1 \\ 2x+3y=10 \end{array} \right\} \begin{array}{l} \text{inconsistent} \\ \downarrow \\ \text{no sol} \end{array}$$

least squares method: $A^T A \hat{x} = A^T b.$ ✓

$$\hat{x} = (A^T A)^{-1} A^T b.$$

26.

27.

28.

29.

30.

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33.

34.

35.